

Scenario 1: CustomerManagement Package

Package Specification

CREATE OR REPLACE PACKAGE CustomerManagement AS

```
PROCEDURE AddCustomer(  
    p_id NUMBER,  
    p_name VARCHAR2,  
    p_dob DATE,  
    p_balance NUMBER  
);
```

```
PROCEDURE UpdateCustomer(  
    p_id NUMBER,  
    p_name VARCHAR2  
);
```

```
FUNCTION GetCustomerBalance(  
    p_id NUMBER  
) RETURN NUMBER;
```

```
END CustomerManagement;  
/
```

Package Body

CREATE OR REPLACE PACKAGE BODY CustomerManagement AS

```
PROCEDURE AddCustomer(  
    p_id NUMBER,  
    p_name VARCHAR2,  
    p_dob DATE,  
    p_balance NUMBER  
)  
AS  
BEGIN  
    INSERT INTO Customers  
    VALUES(p_id,p_name,p_dob,p_balance,SYSDATE);  
  
    COMMIT;  
END AddCustomer;
```

```
PROCEDURE UpdateCustomer(  
    p_id NUMBER,  
    p_name VARCHAR2  
)  
AS
```

```

BEGIN
    UPDATE Customers
    SET Name=p_name,
        LastModified=SYSDATE
    WHERE CustomerID=p_id;

    COMMIT;
END UpdateCustomer;

FUNCTION GetCustomerBalance(
    p_id NUMBER
)
RETURN NUMBER
AS
    v_balance NUMBER;
BEGIN
    SELECT Balance
    INTO v_balance
    FROM Customers
    WHERE CustomerID=p_id;

    RETURN v_balance;
END GetCustomerBalance;

END CustomerManagement;
/

```

Scenario 2: EmployeeManagement Package

Package Specification

```
CREATE OR REPLACE PACKAGE EmployeeManagement AS
```

```

    PROCEDURE HireEmployee(
        p_id NUMBER,
        p_name VARCHAR2,
        p_position VARCHAR2,
        p_salary NUMBER,
        p_department VARCHAR2,
        p_hiredate DATE
    );

```

```

    PROCEDURE UpdateEmployee(
        p_id NUMBER,
        p_salary NUMBER
    );

```

```
FUNCTION CalculateAnnualSalary(  
    p_id NUMBER  
) RETURN NUMBER;  
  
END EmployeeManagement;  
/
```

Package Body

CREATE OR REPLACE PACKAGE BODY EmployeeManagement AS

```
PROCEDURE HireEmployee(  
    p_id NUMBER,  
    p_name VARCHAR2,  
    p_position VARCHAR2,  
    p_salary NUMBER,  
    p_department VARCHAR2,  
    p_hiredate DATE  
)  
AS  
BEGIN  
    INSERT INTO Employees  
    VALUES(  
        p_id,  
        p_name,  
        p_position,  
        p_salary,  
        p_department,  
        p_hiredate  
    );  
  
    COMMIT;  
END HireEmployee;  
  
PROCEDURE UpdateEmployee(  
    p_id NUMBER,  
    p_salary NUMBER  
)  
AS  
BEGIN  
    UPDATE Employees  
    SET Salary=p_salary  
    WHERE EmployeeID=p_id;  
  
    COMMIT;  
END UpdateEmployee;  
  
FUNCTION CalculateAnnualSalary(  
    p_id NUMBER  
)
```

```

RETURN NUMBER
AS
    v_salary NUMBER;
BEGIN
    SELECT Salary
    INTO v_salary
    FROM Employees
    WHERE EmployeeID=p_id;

    RETURN v_salary*12;
END CalculateAnnualSalary;

END EmployeeManagement;
/

```

Scenario 3: AccountOperations Package

Package Specification

```
CREATE OR REPLACE PACKAGE AccountOperations AS
```

```

    PROCEDURE OpenAccount(
        p_accountid NUMBER,
        p_customerid NUMBER,
        p_type VARCHAR2,
        p_balance NUMBER
    );

```

```

    PROCEDURE CloseAccount(
        p_accountid NUMBER
    );

```

```

    FUNCTION GetTotalBalance(
        p_customerid NUMBER
    ) RETURN NUMBER;

```

```

END AccountOperations;
/

```

Package Body

```
CREATE OR REPLACE PACKAGE BODY AccountOperations AS
```

```

    PROCEDURE OpenAccount(
        p_accountid NUMBER,
        p_customerid NUMBER,

```

```

        p_type VARCHAR2,
        p_balance NUMBER
    )
AS
BEGIN
    INSERT INTO Accounts
    VALUES(
        p_accountid,
        p_customerid,
        p_type,
        p_balance,
        SYSDATE
    );

    COMMIT;
END OpenAccount;

PROCEDURE CloseAccount(
    p_accountid NUMBER
)
AS
BEGIN
    DELETE FROM Accounts
    WHERE AccountID=p_accountid;

    COMMIT;
END CloseAccount;

FUNCTION GetTotalBalance(
    p_customerid NUMBER
)
RETURN NUMBER
AS
    v_total NUMBER;
BEGIN
    SELECT SUM(Balance)
    INTO v_total
    FROM Accounts
    WHERE CustomerID=p_customerid;

    RETURN NVL(v_total,0);
END GetTotalBalance;

END AccountOperations;
/

```